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UNIVERSITY OF CHINESE ACADEMY OF SCIENCES

DOCTORAL THESIS

The study of Higgs properties in the four-lepton final state at CMS

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*Dedicated to my family who endured missed birthdays,
missed festivals and countless days alone so that I may
make history*

Chapter 1

Introduction

The Standard Model (SM) of the particle physics gives successful explanation about the constituents of the physical matter and the fundamental forces through which they interact with each other and with composite particles. The electromagnetic, strong and weak forces are described by dedicated theoretical descriptions Quantum Electrodynamics (QED), Quantum Chromodynamics (QCD) and Quantum Flavordynamics (QFD) which are encoded in the SM. Later on, electromagnetic and weak forces among these, were unified as electroweak (EWK) force at the order of 100 GeV unification energy scale. Sheldon Glashow, Abdus Salam, and Steven Weinberg had significant contribution in this milestone. The experimental verifications of the existence of the electroweak interaction were established by the In 1973, existence of neutral currents announced by the Gargamelle collaboration and then in 1981, the discovery of two massive gauge bosons (W and Z) by UA1 and UA2 collaborations verified the existence of EWK force. These boson should be massless (like photon in QED) according to Goldstone's theorem. To overcome this issue, spontaneous symmetry breaking was introduced that generates the masses of the particles by the Higgs mechanism and separates weak interaction from electromagnetic (EM) interaction [**Reference7**, **Reference8**, **Reference9**, **Reference10**, **Reference11**, **Reference12**]. Higgs boson, a new scalar particle, was introduced by the theory being last missing patch of the SM. This triggered the search for this boson of the SM at experimental facilities which lasted for several years. One of the main design and construction motivations of the Large Hadron Collider (LHC) at the CERN was also to look for this new particle given unitary and above the Large Electron-Positron (LEP) constraints, i.e from 114 GeV to 1 TeV. After a long search, the CMS and the ATLAS collaborations of LHC, announced the existence of the heavy scalar boson (the Higgs) on 4 July 2012 [**Reference20**, **Reference21**]. Immediately after the discovery, further tests to measure its properties were begun. Mass of the newly discovered particle, the only free parameter in Higgs sector, has been measured to be $m_H = 125.09 \pm 0.21(stat.) \pm 0.11(syst.)$ GeV [**Reference22**]. Other properties such as spin, parity [**Reference21a**] and couplings of this boson to different particles [**Reference21b**, **Reference21c**] have also been measured and observed to consistent with the predictions of SM. Therefore, the new particle very probable appears as the SM Higgs boson, affiliated with EWK spontaneous symmetry breaking. SM has passed enormous experimental tests in last 50 years and no significant deviation has been seen yet. Experimental evidence of the Higgs

boson is the biggest achievement of the SM. Other success includes the experimental verification of its prior predictions about the W and Z gauge bosons, gluon, top and charm quarks.

Although SM has passed numerous experimental tests with great precision, it is not considered to be theory of everything. There are many phenomena which are not explained by SM i.e Hierarchy problem, neutrino mass and existence of dark matter and dark energy etc. There are several theoretical extensions posed in SM such as the Minimal Super Symmetrical Model (MSSM) and the Next-to-Minimal Super Symmetric Standard Model (NMSSM) or completely new theoretical descriptions, such as string theory and extra dimensions. More precise Higgs boson properties measurements are needed to capture any possible deviation from SM predictions which can be a doorway to new physics.

Model independent production cross section for a particular process is an important property of H boson, where the cross section gives the possibility of the occurrence of that process. In this dissertation, measurements of the integrated or inclusive and differential fiducial cross sections for H boson production in $H \rightarrow 4\ell$ decay channel ($\ell = e, \mu$) in proton-proton collisions at $\sqrt{s} = 13$ TeV are presented. These measurements provide a direct test of perturbative QCD calculations in the Higgs sector of SM at the TeV energy scale. Any deviation of the integrated or the differential cross sections from SM expectations would provide a direct hint to BSM contributions. Nearly similar measurements have already been reported by the ATLAS experiment in $H \rightarrow \gamma\gamma$ [ATLAS:Hgg_8TeV, ATLAS:Hgg_8TeV_13TeV], $H \rightarrow 4\ell$ [ATLAS:H4l_8TeV, ATLAS:H4l_8TeV_13TeV] decay channels, combination of both channels [ATLAS:H4l_Hgg_8TeV, ATLAS:H4l_Hgg_8TeV_13TeV] and by the CMS experiment in $H \rightarrow \gamma\gamma$ decay mode [CMS:Hgg_8TeV].

The Higgs cross section measurement helps to probe the structure, nature of the interaction of H boson with decaying particles. Differential fiducial cross sections are measured using the full Run 2 CMS data recorded at 13 TeV corresponding to several kinematic observables which are sensitive to the H boson production mechanisms: rapidity and transverse momentum of the H boson, associated jet multiplicity and transverse momentum of leading jet. These differential distributions are sensitive to production mechanism of gluon fusion, colliding protons' PDFs, theoretical modeling of hard quarks radiations and relative contributions of different production mechanisms of H boson.

The fragmentation of the dissertation is systematized as follows. An overview of the SM of particle physics, production and decays of H boson are described in Chapter ???. Overview of the LHC and its CMS experiment are presented in Chapter ??. The calibrations and reconstructions of the physics objects within the CMS detector are described in Chapter ??. Chapter 2 presents the procedure of measurements of inclusive and differential fiducial cross sections of H boson production at $\sqrt{s} = 13$ TeV. Finally, the conclusions and outlook of the dissertation are summarized in Chapter ??.

Chapter 2

Probing four lepton final-state for Higgs Cross section measurement

2.0.1 Simulation Samples and Theoretical Predictions

Simulations of the interesting events help us build a realistic model of experiment in high energy. It enables us for a detailed study of physics processes (signal and background). Particularly for proton-proton collisions, generation of simulated events can be separated in many stages. For this analysis, in particular, descriptions of the production of SM Higgs through process of gluon-gluon fusion (Fig. 2.1) are acquired using POWHEG V2 [**powheg**, **powhegvv**] generator. At accuracy level of next-to-next-to-leading order (i.e. NNLO) in perturbative QCD (pQCD), $gg \rightarrow$ process description is obtained by partonic level MC generator NNLOPS.

The POWHEG, provides perturbative QCD calculations at next-to-leading (NLO) accuracy with an connection to parton-shower programs. Using the POWHEG, Higgs production via $gg \rightarrow$ with zero associated jets can be described at accuracy level of NLO. Finite masses of top and bottom quarks has been considered for $gg \rightarrow$ production for all generators. The SM Higgs boson production by

Signal Samples

Descriptions of the SM Higgs boson production are obtained using the POWHEG V2 [**Alioli:2008gx**, **Nason:2004rx**, **Frixione:2007vw**] generator for the five main production modes (ggH , VBF, WH, ZH and ttH) as described in section ?? [**Bagnaschi:2011tu**], [**Nason:2009ai**], [**Hartanto:2015uka**], [**Luisoni:2013kna**]. However, WH and ZH production modes are simulated by the MINLO HVJ extension of the POWHEG [**Luisoni:2013kna**]. Higgs decay to four lepton is simulated by JHUGEN generator [**Gao:2010qx**]. For the WH, ZH and ttH modes, Higgs is allowed to decay to $H \rightarrow ZZ \rightarrow 2\ell 2X$ such that two leptons from 4-lepton events are produced from the decay of associated W, Z bosons or top quarks are included in simulation. Showering of parton-level events is done using PYTHIA8.209, and in all cases matching is performed by allowing QCD emissions at all energies in the shower and vetoing them afterwards according to the POWHEG internal scale. All samples are generated with the NNPDF 3.1 NLO parton distribution functions (PDFs) [**Ball:2014uwa**]. The list of

signal samples with the corresponding cross sections are shown in ???. These samples are also used to assess model dependence of measurement results are indicated in the Table ???.

The MC samples generated by POWHEG and PYTHIA8.209 are used to extract fiducial phase space acceptance and reconstruction efficiencies for individual models separately as explained in Section ???. These samples, with the sample generated by using the NNLOPS with alternate representation of $gg \rightarrow$ process, are utilized for the comparison of measurement results with the theoretical predictions based on SM calculation in Section ???.

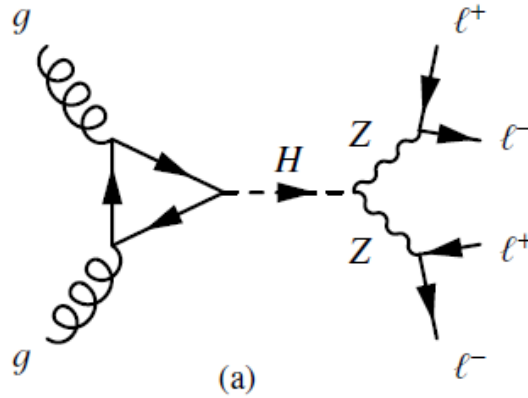


FIGURE 2.1: $gg \rightarrow$ process Feynman diagram.

All SM production modes have been produced individually to approximate the dependence of measurements on underlying assumptions of Higgs production mechanism. The details of the method used to assess the model dependence will be described in Section ???. All these MC samples are produced using POWHEG for the representation of QCD effects at NLO in Higgs production, and by using JHUGEN [Gao:2010qx, Bolognesi:2012mm, Anderson:2013afp] to represent such exotic resonances decays to four-leptons containing all the spin correlations.

Background Samples

For the estimations of the backgrounds using MC samples, the dominant process $q\bar{q} \rightarrow ZZ \rightarrow 4\ell$ was generated at the NLO accuracy level containing s-, t- and u-channel diagrams (Figure 2.2) using POWHEG V2 [Nason:2013ydw] and PYTHIA8, with the same configurations as for the Higgs signal. The dynamical renormalization and factorization scales have been chosen equal to m_{ZZ} since the simulation covers a wide range of invariant masses of ZZ.

The $gg \rightarrow ZZ$ process (Figure 2.2) is simulated at leading order (LO) with MCFM [MCFM, Campbell:2013una]. In order to have matching in spectra of $gg \rightarrow H \rightarrow ZZ$ as predicted at NLO by POWHEG, the showering of these samples has been performed with some different PYTHIA8 settings. This only allows emissions only up to parton-level scale (“wimpy” shower).

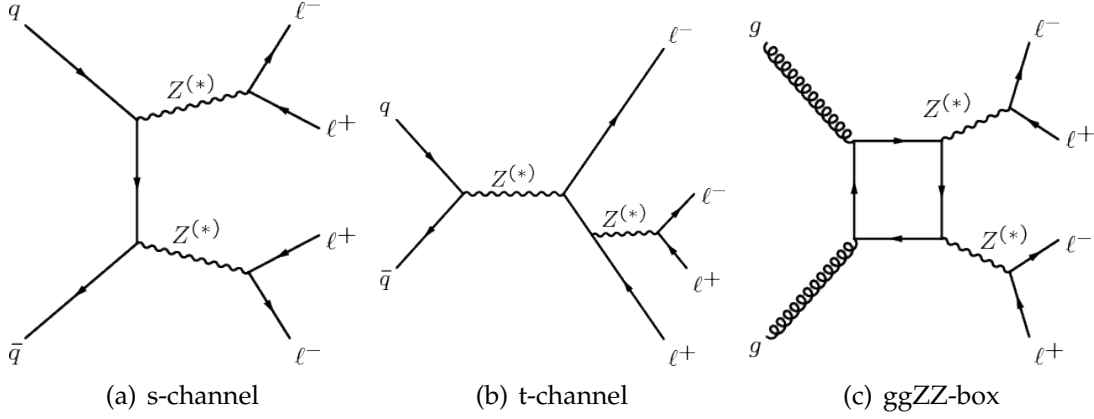


FIGURE 2.2: $q\bar{q} \rightarrow ZZ \rightarrow 4\ell$ process s- and t- channel fymnan diagrams (a) and (b). $gg \rightarrow ZZ \rightarrow 4\ell$ process fymnan diagram (c).

The initial-state and final-state radiations are included for all events generators, which provides hints for the existence of extra (hard) photons in the event. Interference between all amplitudes of process $H \rightarrow 4\ell$ are properly taken into consideration both in POWHEG and the JHUGEN. For the generators (at LO, NLO at NNLO) CTEQ6L [cteq66], CT10 [CT10] and MSTW2008 [MSTW08] PDFs are used.

The generated events were interfaced to PYTHIA 6.4 Tune Z2* [Chatrchyan:2011id] which follows CTEQ6L, to include the fact of the parton shower, underlying events and hadronization.

Processing of all generated events was made via a full simulation for CMS detector using GEANT4 [GEANT] and their reconstruction was done by exploiting same algorithms used for the data analysis. Multiple proton-proton interactions are incorporated in the simulations by reweighting the pileup distribution to match the distribution of the observed data. While measuring the energy of jets and isolation of objects (using FASTJET technique) [Cacciari:2007fd, Cacciari:2008gn, Cacciari:2011ma], the energy deposits from pileup interactions and underlying events are subtracted.

All the data-to-MC scale factors are computed by using a “T&P” technique [CMS:2011lsa], as described in Sections ?? and ??, and are used to correct simulated samples to compensate the differences of leptons’ selection efficiencies in the data and MC simulation. There is negligibe dependence of the selection efficiencies used in the analysis over the primary vertices. More details are presented and discussed in Ref. [Chatrchyan:2013mx].

Pileup Reweighting

For each year, corresponding simulation samples are reweighted to meet pileup distribution in the data. An example of the reweighting procedure for 2018 data is shown in Fig. 2.3.

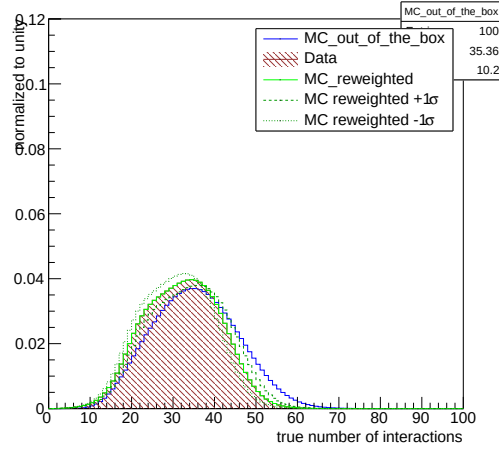


FIGURE 2.3: Pileup distribution in 2018 MC and Data shown before and after the application of PU weights. Up and down variations of 5% in the minimum bias cross section when calculating the weights are also shown.

2.1 Analysis Strategy

In order to minimize the effects of model dependent acceptances, measurements on Higgs cross sections (Inclusive and differential) are made in a fiducial region. Fiducial selection mimics the reconstruction level selection, which is optimized to detect a low mass Higgs (m_H 125 GeV) boson decaying to 4 leptons through a pair of Z bosons. Since in SM there are multiple production modes, the fiducial selection and measurement strategy are designed to be independent of how the Higgs boson is produced.

For this reason, the inclusion of isolation in the fiducial phase space definition is necessary to make the reconstruction efficiencies independent of fiducial phase space.

Furthermore, in sub-leading SM production modes there can be associated vector bosons which can decay to leptons, and therefore the reconstructed Higgs boson candidate may not be composed of the correct combination of the leptons. In order to retain the full information from an unbinned simultaneous fit of the signal and the background PDFs to the four-lepton distribution of invariant mass, it is necessary to treat such “wrong combination” events as a background.

Further signal-induced background can be formed due to imperfect correspondence between quantities at particle level and corresponding quantities at the reconstruction level. Signal events which are from within fiducial phase space but fail the reconstruction are accounted in the fiducial efficiency (ϵ) and the expected contribution from events passing the reconstruction selections but are out of the fiducial phase space (f_{nonfid}) are referred to as “nonfiducial signal” events and are treated as a background, illustrated in Figure 2.4.

The effects of imperfect detector resolution can in general have a non-negligible

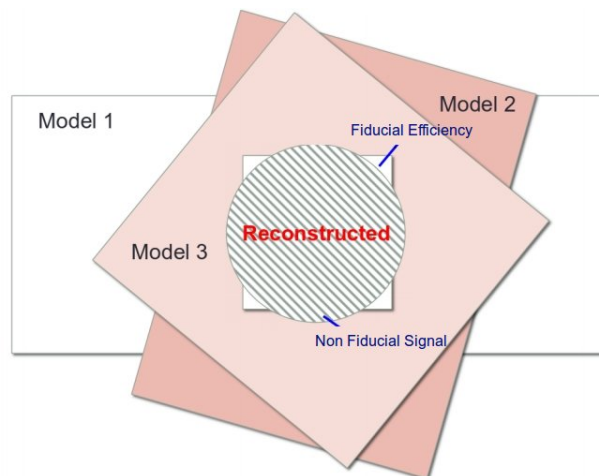


FIGURE 2.4: Fiducial efficiency(ϵ) and nonfiducial(f_{nonfid}) signal definition.

effect on shape of the distribution of measured observables. For this reason, for differential cross section measurements of Higgs, a procedure to correct the detection efficiencies and resolution effects is applied. Throughout this document, this procedure will be referred to as the unfolding procedure, and the unfolded differential distributions will be referred to as distributions at the fiducial level. The reconstruction level selection, fiducial phase space definition, statistical procedure, and their model dependence will be described in the subsequent Sections.

2.1.1 Event Selection and Background Modeling

Event Selection

This dissertation is based on the analysis for which event selection for measurements are similar to what was used in pervious analyses on same topic and in the same final state [HIG-18-001, HIG-16-041, Chatrchyan:2013mxa, Khachatryan:2014kca]. The online event selection is based on having fired High Level Trigger paths as described in [sec:trigpaths]. Unlike in the Run I analysis, the trigger requirement does not depend on the selected final state: it is always the OR of all HLT paths. The reason is in Run II we will be targeting associated production modes that can come with additional leptons, thus improving trigger efficiency further. The events are required to have at least one primary vertex (PV) that fullfills the criteria of having large degree of freedom ($N_{PV} > 4$) with radius of PV being small ($r_{PV} < 2$ cm) and the collisions being restricted along z -axis ($z_{PV} < 24$ cm).

For the double lepton triggers, the requirement on minimum p_T for leading and the subleading lepton are imposed to be 17 and 8 GeV respectively; however, for triple electron triggers, it is 15, 8 and 5 GeV in the same order respectively.

2.1.2 ZZ Candidate Selection

The four-lepton candidates are built from what we call **selected leptons**, which are the tight leptons (defined in sections ?? and ??) that pass the $SIP_{3D} < 4$ vertex constraint and the isolation cuts (defined in sections ?? and ??), where FSR photons are subtracted as briefed in Section ?. A lepton cross cleaning is applied by discarding electrons which are within $\Delta R < 0.05$ of selected muons.

The construction and selection of four-lepton candidates proceeds according to the following sequence:

1. **Z candidates** are built from oppositely charged and same flavoured leptons (e^+e^- , $\mu^+\mu^-$) with additional requirement on dilepton mass as $12 < m_{\ell\ell(\gamma)} < 120 \text{ GeV}$. Z boson candidate is also required to include FSR photons (if any).
2. **ZZ candidates** are then built from non-overlapping Z boson candidates. The one who is reconstructed with $m_{\ell\ell}$ closer to the SM Z boson mass is regarded as Z_1 and other as Z_2 . Additional requirements on ZZ candidates are as under:
 - **Ghost removal** : Selected four leptons are required to have $\Delta R(\eta, \phi) > 0.02$ among each other.
 - **lepton p_T** : Among selected four leptons, two are required to fulfill $p_{T,i} > 20$ and $p_{T,j} > 10$.
 - **QCD suppression**: Dilepton invariant mass is required to pass (regardless of flavor) $m_{\ell\ell} > 4$ for QCD induced J/Ψ veto.
 - **Z_1 mass**: $40 \text{ GeV} < m_{Z_1} < 120 \text{ GeV}$
 - **four-lepton invariant mass**: $m_{4\ell} > 70$
3. Events containing one selected ZZ candidate at least should form the **signal region**.

2.1.3 Choice of best ZZ Candidate

Unlike in the Run I analysis, the best ZZ candidate is now chosen after all kinematic cuts, a change that allows to test other selection strategies for this candidate choice. This is especially relevant for events with more than four selected leptons, having in mind search for Higgs boson production modes where associated particles can decay to leptons, such as VH and ttH.

For fiducial cross section measurement, Z_1 is chosen with mass closest to the mass of SM Z boson mass while Z_2 is chosen from Z candidates whose lepton pair gives higher p_T sum.

Reconstruction of the jets is incumbent to explore the differential cross section measurements with respect to jets related observables. Jets are reconstructed with $p_T \geq 30 \text{ GeV}$ within $|\eta| \leq 4.7$, discussed in Section ?. To remove overlapping of jets overlap from leptons or selected photons, it is required to have at least $\Delta R > 0.5$ where $\Delta R \equiv \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$ [Chatrchyan:2013mx] with respect to any of the selected leptons and FSR photons candidates.

Background Modeling

The dominant and sub-dominant irreducible backgrounds for $\rightarrow 4\ell$ process are the production of ZZ via annihilation of a $q\bar{q}$ pair and the gg fusion respectively. Other background is reducible which is of the instrumental nature and arises from the process where partial jets are misidentified as leptons. The main processes which contributes in this background are: production of jets in association with Z boson, production of the jets in association with WZ boson pair and the $t\bar{t}$ pair production. This background is as Z+X.

TABLE 2.1: Number of observed candidates, expected background and the signal events passing all selections in mass range $m_{4\ell} > 70$ GeV corresponding to three different years. Z+X is estimated by control regions in the data while ZZ and signal events are estimated using MC simulation.

Channel	4e	4 μ	2e2 μ	4 ℓ
2016 (13 TeV)				
$q\bar{q} \rightarrow ZZ$	$192.7^{+18.6}_{-20.1}$	$360.2^{+24.9}_{-27.3}$	$471.0^{+32.6}_{-35.7}$	$1023.9^{+68.9}_{-76.0}$
$gg \rightarrow ZZ$	$41.2^{+6.3}_{-6.1}$	$69.0^{+9.5}_{-9.0}$	$101.7^{+14.0}_{-13.3}$	$211.8^{+28.9}_{-27.5}$
Z + X	$21.1^{+8.5}_{-10.4}$	$34.4^{+14.5}_{-13.2}$	$59.9^{+27.1}_{-25.0}$	$115.4^{+31.9}_{-30.1}$
Sum of backgrounds	$255.0^{+23.9}_{-25.1}$	$463.5^{+31.9}_{-33.7}$	$632.6^{+44.2}_{-46.1}$	$1351.1^{+85.8}_{-91.2}$
Signal ($m = 125$ GeV)	$12.1^{+1.3}_{-1.4}$	23.8 ± 2.1	30.3 ± 2.6	66.3 ± 5.6
Total expected	$267.1^{+24.9}_{-26.1}$	$487.4^{+33.1}_{-34.9}$	$662.9^{+45.7}_{-47.5}$	$1417.4^{+89.1}_{-94.3}$
Observed	293	505	681	1479
2017 (13 TeV)				
$q\bar{q} \rightarrow ZZ$	$233.86^{+y.y}_{-z.z}$	$441.08^{+y.y}_{-z.z}$	$569.51^{+y.y}_{-z.z}$	$1244.45^{+y.y}_{-z.z}$
$gg \rightarrow ZZ$	$49.11^{+y.y}_{-z.z}$	$81.50^{+y.y}_{-z.z}$	$121.05^{+y.y}_{-z.z}$	$251.66^{+y.y}_{-z.z}$
Z + X	$12.39^{+y.y}_{-z.z}$	$36.39^{+y.y}_{-z.z}$	$44.45^{+y.y}_{-z.z}$	$93.24^{+y.y}_{-z.z}$
Sum of backgrounds	$295.37^{+y.y}_{-z.z}$	$558.97^{+y.y}_{-z.z}$	$735.02^{+y.y}_{-z.z}$	$1589.35^{+y.y}_{-z.z}$
Signal ($m = 125$ GeV)	$13.85^{+y.y}_{-z.z}$	$28.64^{+y.y}_{-z.z}$	$35.55^{+y.y}_{-z.z}$	$78.04^{+y.y}_{-z.z}$
Total expected	$309.22^{+y.y}_{-z.z}$	$587.60^{+y.y}_{-z.z}$	$770.57^{+y.y}_{-z.z}$	$1667.39^{+y.y}_{-z.z}$
Observed	307	602	797	1706
2018 (13 TeV)				
$q\bar{q} \rightarrow ZZ$	333^{+57}_{-53}	622^{+31}_{-44}	815 ± 73	1770^{+98}_{-101}
$gg \rightarrow ZZ$	$75.1^{+14.3}_{-13.5}$	$116.6^{+11.7}_{-12.8}$	176.9 ± 23.0	$368.5^{+29.5}_{-29.6}$
Z + X	19.3 ± 7.2	50.8 ± 15.2	64.6 ± 15.6	134.7 ± 22.9
Sum of backgrounds	$428^{+59.2}_{-55.2}$	$790^{+36.4}_{-48.3}$	1057 ± 78.1	$2274^{+104.9}_{-107.7}$
Signal ($m = 125$ GeV)	$19.6^{+3.3}_{-3.1}$	$40.8^{+2.5}_{-2.9}$	50.7 ± 5.6	$111.1^{+6.9}_{-7.0}$
Total expected	$447^{+59.3}_{-55.2}$	$830^{+36.5}_{-48.4}$	1108 ± 78.3	$2385^{+105.1}_{-107.9}$
Observed	462	850	1130	2442

For fiducial cross section measurement for $\rightarrow 4\ell$, both of irreducible backgrounds ($q\bar{q} \rightarrow ZZ$ and $gg \rightarrow ZZ$) are estimated from Monte Carlo (MC) simulation as done earlier in Ref. [Chatrchyan:2013mx]. In addition, following the study in

Ref. [Bonvini:2013], in these measurements the expected yield for the $gg \rightarrow ZZ$ background has been increased by assigning the same ($m_{4\ell}$ dependent) k-factor as signal $\rightarrow 4\ell$ process to leading order (LO) cross section of background. In the region ($105 < m_{4\ell} < 140$ GeV) this factor leads to an increase of expected $gg \rightarrow ZZ$ background by factor of ~ 2.66 . After inclusion of this k-factor, the $gg \rightarrow ZZ$ process is still a sub-dominant background in all measurements.

Reducible background (Z+X) is estimated using data. A tight-to-loose “fake rate” technique is used in the control regions of data by following Ref. [Chatrchyan:2013mx]. Z+X background yields are determined using two methods:

- Opposite Sign (OS) method
- Same Sign (SS) method

Both methods are found to be consistent with each other within the statistical uncertainties. First control region is attained with the requirement that two leptons do pass loose selections which does fulfill the criteria of Z_1 candidate (failing ID and isolation requirements), while the other two leptons in the event passes the criteria of Z_1 candidate. Thus, the sample is represented as “2 Prompt + 2 Fail” (2P2F) which is populated with the events which contain two prompt leptons (which are mostly Drell-Yan with small fraction of Z/γ^* and $t\bar{t}$ events) only. Second control region is attained by requiring that three leptons pass the final requirements. It is represented as “3 Prompt + 1 Fail” (3P1F) sample.

The reducible background’s $m_{4\ell}$ shape is attained from the OS method by performing a fit of $m_{4\ell}$ distributions of 3P1F and 2P2F events independently by empirical analytical functional constructed by Landau [landau:1944] in Ref. [Chatrchyan:2013mx]. Reducible background prediction in all three channels fitted together using this with total uncertainty indicated is shown in Figure ???. For differential measurements in $\rightarrow 4\ell$, shape of reducible background is attained from templates built from the control regions in the data for each differential bin of the observable considered for measurement. This procedure is similar to what was used for spin-parity studies as described in the Ref. [Chatrchyan:2013mx]. More details on this procedure of differential measurements of $\rightarrow 4\ell$ will be given in Section ??.

The number of estimated signal and the background events, along with observed events in data passing all selections in mass range $m_{4\ell} > 70$ GeV are listed in Table 2.1 for full Run 2 datasets.

Reconstructed four-leptons invariant mass distribution for whole Run 2 datasets and MC samples in full and high mass range is shownn in the Figure 2.5.

2.1.4 Definition of the Fiducial Volume

Fraction of all $\rightarrow 4\ell$ events that get selected following the event selection procedure can change significantly for various production mechanisms of Higgs and for its distinct exotic models. In order to minimize the effects of model dependent acceptances, measurements on Higgs cross sections (Inclusive and differential) in $\rightarrow 4\ell$ are made in a fiducial region. This fiducial phase space is defined so as to

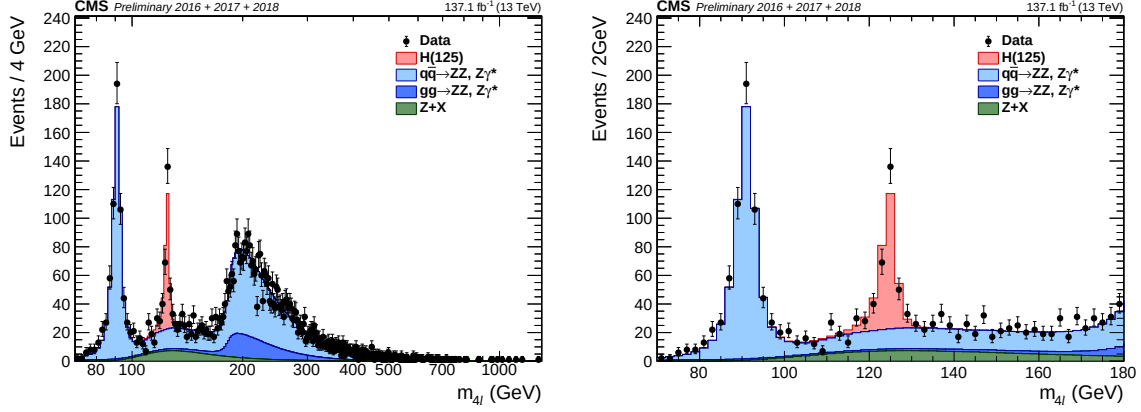


FIGURE 2.5: Invariant mass distributions of invariant mass of four-leptons using full Run 2 datasets in the full (left) and the low (right) mass range, along with predictions for Standard Model Higgs ($m_H = 125.0$ GeV), including $Z \rightarrow 4\ell$ resonance.

closely match with the experimental acceptance as defined from selections made at reconstruction level.

Requirement of fiducial phase space is minimum four leptons having $p_T > 7$ GeV and $|\eta| < 2.5$ in case of electron, and $p_T > 5$ GeV and $|\eta| < 2.4$ in case of muon. Besides, the leading and sub-leading leptons are required to have $p_T > 20$ GeV and $p_T > 10$ GeV respectively. The four-momenta of each lepton is calculated at the hard scattering level (PYTHIA status code = 3), before any final state radiation occurs. Each lepton isolation is calculated by adding p_T of all stable particles (PYTHIA status code = 1) within a cone size $\Delta R < 0.3$ around the lepton. All neutrinos and FSR photons are excluded from above summation. Then the lepton's relative isolation which is defined by ratio of this sum and p_T of considered lepton and is required to be less than 0.35, consistent with reconstruction level isolation requirement [Chatrchyan:2013mx]. The requirement of isolation in definition of fiducial phase space is crucial to control the difference of signal selection efficiency among different models. The studies based on simulation samples show that signal efficiency can change up to 45% among different models without requiring lepton isolation at fiducial level as shown in Table ??, the maximum difference comes from $\gamma\gamma$ and $gg \rightarrow$ production modes. This behavior is expected as $\gamma\gamma$ events has more hadronic activity around leptons as compared to $gg \rightarrow$ events, so the cut at relative isolation kills more events in $\gamma\gamma$ than $gg \rightarrow$.

Furthermore, fiducial phase space is defined by several other topological and kinematics selections. There must be two SFOS lepton pairs where Z_1 being the one with invariant mass closest to the SM Z boson and reconstructed with $40 \leq m(Z_1) \leq 120$ GeV. Z_2 is reconstructed by remaining SFOS pair of leptons within inv. mass range of $12 \leq m(Z_2) \leq 120$ GeV. In case of more than one Z_2 passing the criteria, leptons pair with highest $\sum |p_T|$ is selected. To remove the overlapping between leptons, we ask for $\Delta R(\ell_i \ell_j) > 0.02$ for any $i \neq j$. To remove contamination from QCD low masses resonance, any opposite-charge pair of leptons is required to meet the condition $m(\ell^+ \ell'^-) \geq 4$ GeV. Finally, We require that all selected leptons

must emerge from Higgs boson decay and invariant mass of four-leptons must fulfill $105 \text{ GeV} < m(4\ell) < 140 \text{ GeV}$. Demand for $m(4\ell)$ helps to control the contamination from off-shell gg fusion production which is of considerable size, capable to increase the total cross section by a few percent [Kauer:2013cga, Kauer:2012hd]. The requirements that defines fiducial phase space are also gathered in Table ??.

TABLE 2.2: Overview of all requirements utilized to define the fiducial phase space.

Requirements on $\rightarrow 4\ell$ fiducial phase space	
Lepton kinematics and the isolation	
p_T of the leading lepton	$p_T > 20 \text{ GeV}$
p_T of sub-leading lepton	$p_T > 10 \text{ GeV}$
Additional electrons (muons) p_T	$p_T > 7 \text{ (5) GeV}$
Pseudorapidity of electrons (muons)	$ \eta < 2.5 \text{ (2.4)}$
Sum of the scalar p_T of all the stable objects within $\Delta R < 0.3$ from the lepton	$< 0.35 p_T$
Event topology	
From leptons satisfying criteria above, minimum two SFOS lepton pairs	
Invariant mass of Z_1 candidate	$40 < m(Z_1) < 120 \text{ GeV}$
Invariant mass of Z_2 candidate	$12 < m(Z_2) < 120 \text{ GeV}$
Separation between selected four leptons	$\Delta R(\ell_i \ell_j) > 0.02$
Invariant mass of any OS lepton pair	$m(\ell_i^+ \ell_j^-) > 4 \text{ GeV}$
Invariant mass of selected four leptons	$105 < m_{4\ell} < 140 \text{ GeV}$

It has been established based on simulation samples that with including isolation for the events passing fiducial phase space definition reconstruction efficiencies faintly depends on Higgs properties and its production modes. The remaining model dependence will be quoted as a separate uncertainty as explained in Section ?. Part of the signal events passing fiducial phase space definition ($\mathcal{A}_{\text{fid}}$), and reconstruction efficiencies (ϵ) for SM production modes are registered in Table ?.

TABLE 2.3: Summary of different SM signal models. The MC samples are from full Run 2 yearly productions (2016, 2017 and 2018) assuming $m_H = 125.0$ GeV. The fiducial acceptance ($\mathcal{A}_{\text{fid}}$) defined the fraction of events passing fiducial phase space selections, reconstruction efficiency (ϵ) represented by the ratio of reconstructed to accepted events. Also, f_{nonfid} which are reconstructed events but does not pass fiducial phase space definition at generator level. Values are obtained using 13 TeV signal models and the uncertainties shown are only statistical uncertainties.

Signal process	$\mathcal{A}_{\text{fid}}$	ϵ	f_{nonfid}	$(1 + f_{\text{nonfid}})\epsilon$
Individual production of Higgs boson				
gg $\rightarrow$ H (POWHEG)	0.402 ± 0.001	0.592 ± 0.002	0.053 ± 0.001	0.624 ± 0.002
VBF (POWHEG)	0.444 ± 0.002	0.605 ± 0.003	0.043 ± 0.001	0.631 ± 0.003
WH (POWHEG+MINLO)	0.325 ± 0.002	0.588 ± 0.003	0.075 ± 0.002	0.632 ± 0.004
ZH (POWHEG+MINLO)	0.340 ± 0.003	0.594 ± 0.005	0.081 ± 0.004	0.643 ± 0.006
ttH (POWHEG)	0.314 ± 0.003	0.585 ± 0.006	0.169 ± 0.006	0.684 ± 0.007

Appendix

